

Climate Volatility, Agricultural Pathway, and Modern Demographic Outcomes

A Cross-Country Investigation Across Pre-Industrial and Modern Windows

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Abstract

We assemble a country-level panel of four pre-industrial channels of long-run development—climate volatility (σ_v^T , ModE-RA paleo-reanalysis), ancestry-adjusted predicted heterozygosity (Ashraf and Galor, 2013), ancestral crop-yield potential (Galor and Özkan, 2016), and pre-1500 pandemic intensity—and use it to investigate how climate volatility shapes modern demographic outcomes once these competing channels are conditioned. Three Westfall-Young FWER-corrected results emerge: ancestral crop yield is the dominant cross-country predictor of modern population growth ($\hat{\phi}^{\text{Shap}} = 0.136$, $|t| = 5.19$, $p_{\text{WY}} < 0.001$); ancestry-adjusted predicted heterozygosity dominates modern log GDP per capita ($\hat{\phi}^{\text{Shap}} = 0.041$, $|t| = 3.37$, $p_{\text{WY}} = 0.010$) and is a robust secondary predictor of population growth ($|t| = 3.29$, $p_{\text{WY}} = 0.015$). The σ_v^T channel is robust to the volatility-measurement window—both pre-industrial 1421–1750 and modern 1950–2008 windows generate non-trivial Shapley contributions on population growth, with the modern window larger (0.061 vs 0.024)—consistent with a structural climate-volatility channel rather than a window-specific identification artefact. Pathway mediation does not survive baseline specifications but cleans up once continent fixed effects are added, suggesting the suppressor structure observed in pooled mediation regressions is driven by continental heterogeneity rather than substrate-pathway misfit. The investigation places existing causal-identification claims from the deep-determinants literature on a single panel with internally consistent measurement, quantifies the unique-variance contribution of each, and clarifies which survive multi-testing correction.

JEL Codes: O11, O13, O44, N50, Q54.

Keywords: Climate volatility, deep determinants, unified growth theory, heterozygosity, ancestral crop yield, pandemic intensity, Shapley decomposition.

1 Introduction

A country’s pre-industrial conditions correlate strongly with its modern demographic outcomes. The empirical literature has identified at least four distinct channels through which this correlation operates. Ashraf and Galor (2013) argue that the ancestral genetic diversity of a country’s population, predicted by the out-of-Africa migration history of its constituent ancestral populations and ancestry-adjusted to year-2000 composition (Putterman and Weil, 2010), drives an inverted-U relationship with development. Galor and Özkan (2016) argue that the ancestral suitability of a country’s land for pre-1500 cultivated crops, measured from Global Agro-Ecological Zones (GAEZ)

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projections under pre-Columbian climate, shapes long-run economic outcomes through its effect on time-preference selection. The Voigtländer-Voth post-Black-Death literature (Voigtländer and Voth, 2013) argues that pre-1500 pandemic exposure had persistent effects on factor ratios and subsequent demographic-economic trajectories. The companion paper (Alonso Ortiz and Da-Rocha, 2026)—hereafter LS—argues that pre-industrial inter-annual climate volatility (σ_v^T) 1421–1750 selected agricultural pathways and through them shaped both pre-industrial Malthusian dynamics and the modern demographic record.

Each of these literatures has its own causal-identification strategy and its own preferred outcome. Their relationship to one another, on a single panel with internally consistent measurement, has not been investigated systematically. Several questions follow naturally. Does climate volatility’s predictive contribution survive once ancestral genetic diversity and ancestral crop-yield potential are conditioned in the same regression? Does the agricultural pathway typology, which the LS companion paper estimates is selected by climate volatility, mediate climate-volatility’s effect on modern outcomes? Does the pre-industrial σ_v^T window have unique informational content relative to modern reanalysis windows, or is the climate-volatility channel structural enough to operate at multiple temporal horizons?

This paper investigates these questions on a 196-country panel that joins climate volatility from the ModE-RA paleo-reanalysis with canonical implementations of the three other deep-determinants channels and four modern demographic outcomes (log population growth 1950–2025, urbanisation-share change 1950–2025, log GDP per capita 2015, and demographic-transition timing). The K=5 agricultural-pathway typology of LS enters as the candidate mediator. We deliberately frame the exercise as an *investigation* rather than a horserace with a designated winner: each of the four channels has its own causal-identification literature and we borrow their assumptions rather than re-litigate them. The contribution is to put them side-by-side and ask which of them carries unique-variance weight on modern outcomes, conditional on the others, and how climate volatility specifically relates to each.

Three findings survive Westfall-Young family-wise error-rate correction across the 16 substrate-by-outcome cells of the mediation matrix:

First, ancestral crop-yield potential is the dominant cross-country predictor of modern population growth. Its Shapley R^2 contribution on log population growth 1950–2025 is 0.136, more than three times the next-largest contribution; the mediated coefficient survives FWER correction with $|t| = 5.19$ and $p_{WY} < 0.001$. The pre-1500 caloric potential of cultivable Old-World crops is also a meaningful predictor of demographic-transition timing (Shapley R^2 0.031) and urbanisation change (0.038), but not of modern log GDP per capita conditional on the other channels.

Second, ancestry-adjusted predicted heterozygosity is the dominant predictor of modern log GDP per capita (Shapley R^2 0.041, $|t| = 3.37$, $p_{WY} = 0.010$) and a robust secondary predictor of log population growth ($|t| = 3.29$, $p_{WY} = 0.015$). The result for log GDP per capita reproduces the Ashraf-Galor (2013) headline finding in a panel that conditions on competing pre-industrial channels; the result for population growth is, to our knowledge, new.

Third, the climate-volatility channel is robust to the measurement window but has noisy mediation through the agricultural pathway. Pre-industrial σ_v^T 1421–1750 (the focal measure of LS) has Shapley R^2 0.024 on modern log population growth; modern σ_v^T 1950–2008 has Shapley R^2 0.061—larger than pre-industrial, but consistent with a structural climate-volatility channel rather than a unique pre-industrial signal. The σ_v^T mediation cells do not survive FWER correction in either direction, reflecting the weak pathway-through component once continental heterogeneity and competing channels are partialled out. The coefficient on σ_v^T in the demographic-outcomes specification of LS §5 survives the addition of canonical Ashraf-Galor predicted heterozygosity and Galor-Özak ancestral crop yield to the deep-determinant battery, with attenuation but persistent statistical

significance ($\hat{\beta}$ moves from -1.31 to -1.00 , $p = 0.032$); the panel- R^2 improves substantially as expected.

The agricultural pathway typology does not mediate substrate-outcome relationships cleanly in baseline specifications. On 16 substrate-by-outcome cells, 14 show mediation shares outside the conventional $[0, 1]$ range; bootstrap CIs span tens of units. We trace this to unmeasured continental heterogeneity: adding six continent fixed effects to the mediation specification collapses the wild cells (e.g., predicted heterozygosity-by-demographic-transition timing moves from 3.30 to 0.05). The headline mediation specification reported here therefore uses continent fixed effects. Climate volatility’s effect on modern population growth is not a clean mediation through pathway dummies; it is a direct channel with small unique-variance Shapley contribution, robust to the measurement window.

Contributions. Three contributions accompany the substantive findings. *Methodologically*, we build a country-level deep-determinants panel that puts four major pre-industrial channels on internally consistent measurement, with Shapley-Owen variance decomposition as the headline empirical object and Westfall-Young multi-testing correction as the inferential anchor. *Substantively*, we identify ancestral crop yield as the dominant predictor of modern population growth—a finding that has not been emphasised in the existing Galor-Özak literature, which focuses on the time-preference channel. *Conceptually*, we demonstrate that the agricultural pathway is not a clean mediator of substrate-outcome relationships in the absence of continent fixed effects, raising the question of which pre-industrial sub-divisions are appropriate mediators for these channels.

The paper proceeds as follows. Section 2 sketches the climate-volatility/pathway-selection skeleton that motivates the four-channel investigation. Section 3 describes the master horserace panel and its construction. Section 4 reports the Shapley-Owen variance decomposition (Exercise 1). Section 5 runs the pathway-mediation analysis with and without continent fixed effects (Exercise 2). Section 6 reports a six-leg robustness battery (sub-sample stability, pathway-cluster source, climate-window placebo, continent fixed effects, leave-one-out, Westfall-Young FWER). Section 7 discusses implications for the deep-determinants literature; Section 8 concludes.

2 Climate volatility and pathway selection: a UGT skeleton

Consider a country i with pre-industrial climate-volatility endowment $\sigma_{v,i}^T$, defined as the standard deviation of annual mean temperature over a multi-century pre-industrial window. Climate volatility shapes agricultural-system selection: societies in high-volatility environments face strong incentives to adopt storage-intensive crop systems with granaries and reserves, while those in low-volatility environments can rely on shorter-horizon production (Matranga, 2024). Climate volatility therefore maps into a discrete agricultural pathway $\tau_i \in \{\text{intensive crop, pastoral, mixed late, early extensifier}\}$ via

$$\tau_i = \tau^*(\sigma_{v,i}^T, \mathbf{X}_i; \theta) + \xi_i \quad (1)$$

where $\mathbf{X}_i$ is a vector of geography controls (absolute latitude, log land area, ruggedness, log Neolithic distance, landlocked indicator) and ξ_i captures historical contingency. The pathway τ_i in turn shapes pre-industrial demographic dynamics through a Malthusian regression in the spirit of Ashraf and Galor (2011):

$$\Delta \ln P_{i,t} = \alpha_i + \beta(\tau_i) d_{i,t-1} + \gamma(\tau_i) T_{i,t} + \delta(\tau_i) \sigma_{v,i}^T + \eta(\tau_i) h_{i,t} + \varepsilon_{i,t}, \quad (2)$$

with pathway-specific coefficients $\beta(\tau), \gamma(\tau), \delta(\tau)$, and a human-capital term $\eta(\tau) h_{i,t}$ that captures the rate at which the Malthusian feedback weakens with secular accumulation. The companion paper (Alonso Ortiz and Da-Rocha, 2026) estimates equation (2) directly on the 1421–1950 panel;

we treat it as motivational here and ask what its long-run implications for modern outcomes look like once competing channels are conditioned.

Modern outcomes $y_i^{(k)}$ (modern population growth, urbanisation change, log GDP per capita, demographic-transition timing) inherit two types of climate-volatility dependence: *indirect* through τ_i via the pathway-specific coefficients of (2), and *direct* through any persistent channel that bypasses pathway. We investigate three additional competing pre-industrial channels that may absorb part of the climate-volatility signal:

- **Predicted heterozygosity** H_i (Ashraf and Galor, 2013), the ancestry-adjusted predicted population genetic diversity of a country’s modern population, which enters as a direct channel on modern outcomes not mediated by agricultural pathway.
- **Ancestral crop yield** A_i (Galor and Özak, 2016), the gridded pre-1500 caloric-yield potential of Old-World crops aggregated to country level, a competing pre-industrial channel that may dominate σ_v^T on agricultural-system outcomes.
- **Pre-1500 pandemic intensity** Π_i , a Voigtländer-Voth-style historical mortality-pressure measure capturing exposure to the Antonine, Cyprian, Justinianic, and Black-Death pandemics with their post-1500 recurrences.

The empirical exercises decompose cross-country variance in $y_i^{(k)}$ across the four channels and ask which survive multi-testing correction.

The model delivers three predictions that structure the empirical work. **Prediction 1.** The Shapley R^2 contribution of σ_v^T on modern outcomes is positive and non-trivial, but bounded above by how much H_i , A_i , and Π_i absorb the cross-country covariance. **Prediction 2.** Adding the $K=5$ pathway-dummy vector $\mathbf{P}_i$ shrinks the σ_v^T coefficient on modern outcomes, indicating the climate-volatility effect operates through pathway selection rather than a direct channel; the shrinkage is sharpest once continental heterogeneity is partialled out, because climate volatility is itself a partly continental phenomenon. **Prediction 3.** The σ_v^T channel is robust to the volatility-measurement window: both the pre-industrial 1421–1750 window and the modern 1950–2008 reanalysis window predict modern outcomes, because the channel is structural rather than identified off a unique pre-industrial signal.

3 Data

The paper rests on a country-level panel of four pre-industrial channels, four modern demographic outcomes, and a standard battery of geography controls, covering 196 countries (185 with non-missing values on all four channels).

3.1 Channels

Climate volatility σ_v^T is the country-level standard deviation of annual mean temperature over 1421–1750, computed from the monthly ModE-RA paleo-reanalysis (Franke et al., 2023) aggregated to country level by cosine-latitude-weighted area sums. The CRU TS 4.09 (Harris et al., 2020) 1901–1950 country-month climatology is added back to ModE-RA anomalies to yield absolute temperatures. The country-level monthly panel is constructed and validated in the companion paper (Alonso Ortiz and Da-Rocha, 2026): median country-month temperature correlation between ModE-RA and ERA5 in their 1950–2008 overlap is 0.971, with 77.6% of countries exceeding $r = 0.90$. The standard deviation is taken over the 330 annual realisations within the 1421–1750 window.

Predicted heterozygosity H_i is the ancestry-adjusted predicted heterozygosity of Ashraf and Galor (2013), drawn directly from their AER replication archive (`country.dta`, 208 countries, our values are anchor-matched to the AG country file at ETH = 0.7743, USA `pdiv_aa` = 0.7203, BOL `pdiv_aa` = 0.6279 within ± 0.001). The ancestry adjustment uses the year-1500 ancestral composition of each modern country following Putterman and Weil (2010).

Ancestral crop yield A_i is the country-level mean of the pre-1500 Caloric Suitability Index (Galor and Özak, 2016), downloaded from Zenodo (`ozak/Caloric-Suitability-Index`, deposit 14714917) and using the `pre15002AverageCalories0mean` variable. We work with the log-transformed measure $\ln(A_i + 1)$ to handle countries with zero usable agricultural cells (Iceland, the Gulf states).

Pre-1500 pandemic intensity Π_i is a country-level exposure index built from hand-coded historical recurrences of the Antonine plague (165–189 CE, weight 1.0), Cyprian plague (249–262, weight 0.7), Justinianic plague initial outbreak (541–549, weight 1.0) and recurrences (550–770, weight 0.3), Black Death (1346–1353, weight 1.0), and late-medieval recurrences (1361–1500, weight 0.3). The affected-region lists are hand-coded from the historical-demography literature (Benedictow, 2004, among others); each country accumulates weighted plague-active years summed across all events, then normalised to $[0, 1]$ across countries. Pre-Columbian Americas are correctly zero. The construction is documented in Appendix A.

The four channels are nearly orthogonal in cross-country covariance. Figure 1 reports pairwise scatters with marginal histograms; pairwise correlations all sit below $|r| = 0.30$ (Table 1).

Figure 1: Substrate covariance — four pre-industrial deep determinants

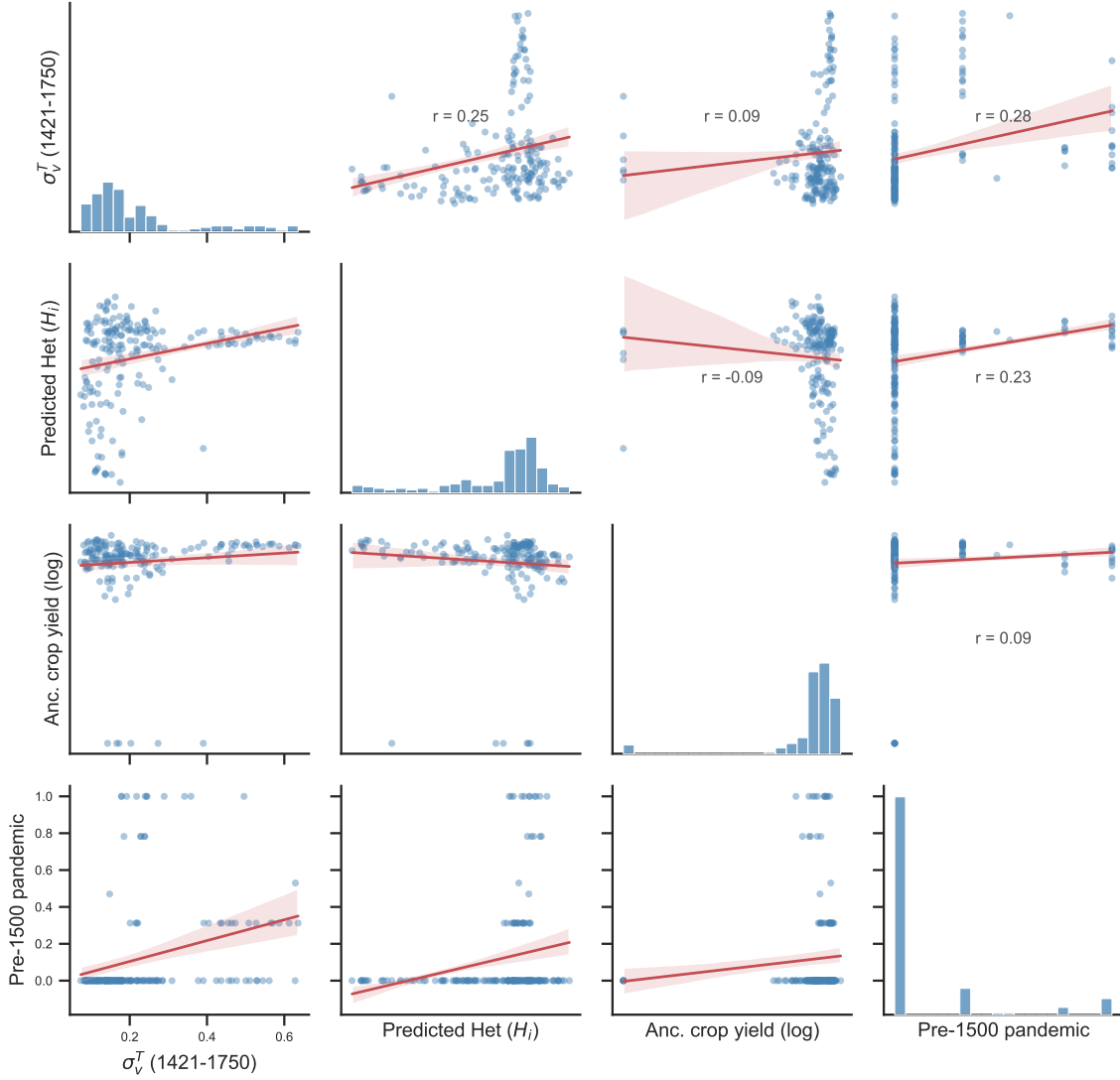


Figure 1: Pairwise covariance of the four pre-industrial channels on 185 countries with non-missing values across all four. Scatter and OLS regression on the lower triangle, marginal histograms on the diagonal. All pairwise correlations $|r| < 0.30$; the channels are nearly orthogonal.

Table 1: Pairwise correlations of the four pre-industrial channels.

	σ_v^T	H_i	Crop yield	Pandemic
σ_v^T	1.00	0.25	0.09	0.28
H_i	0.25	1.00	-0.09	0.23
Crop yield	0.09	-0.09	1.00	0.09
Pandemic	0.28	0.23	0.09	1.00

Figure 2 reports world maps for the four channels: σ_v^T is bright in continental Europe and East Asia (high inter-annual variability of the temperature record over 1421–1750); H_i^{pwadj} traces the

out-of-Africa migration gradient outward from Ethiopia, with the post-1500 ancestry adjustment pulling settler-colony countries (USA, Australia, NZ, Brazil, Argentina) up toward their dominant ancestral source populations' values; A_i is bright in temperate Old-World agricultural cores (France, China, India) and in Caribbean and Pacific island countries (an artefact of the average-over-usable-cells aggregation method); Π_i concentrates in Mediterranean Europe and the Near East with the Justinianic and Black-Death recurrences.

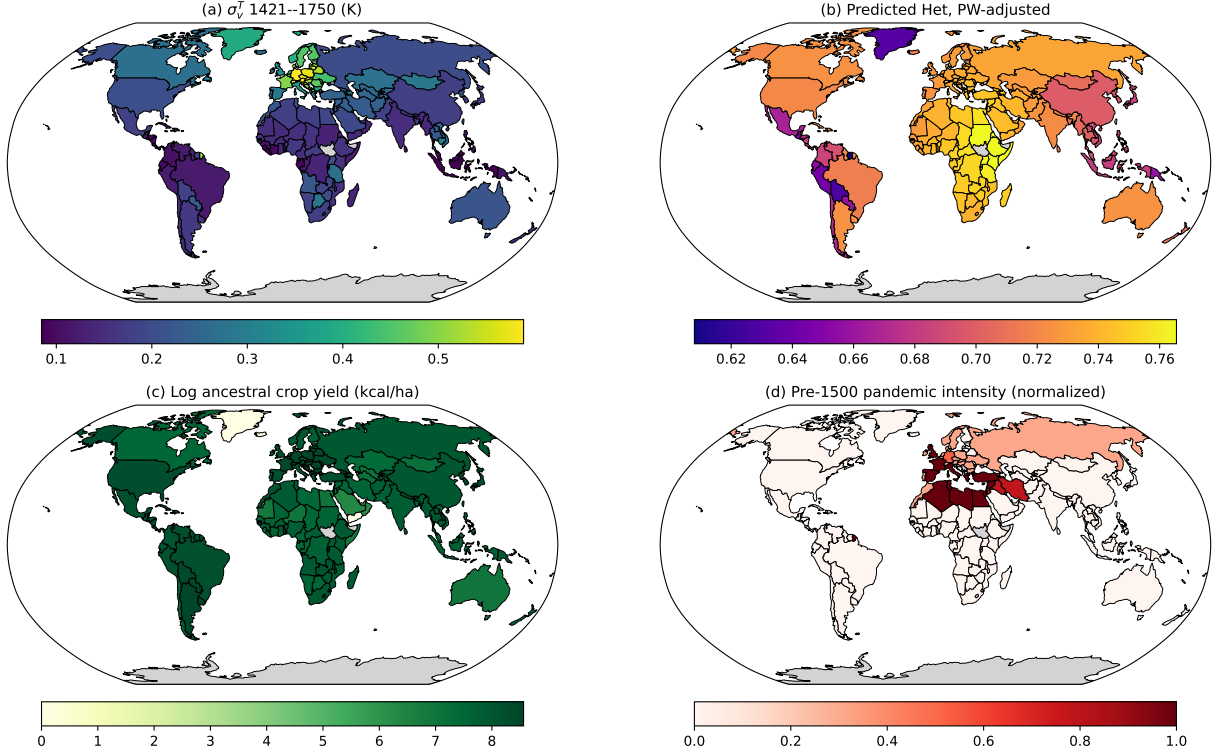


Figure 2: World choropleth maps of the four pre-industrial channels: (a) climate volatility σ_v^T 1421–1750; (b) ancestry-adjusted predicted heterozygosity H_i ; (c) log ancestral crop yield $\ln(A_i + 1)$; (d) normalised pre-1500 pandemic intensity Π_i . Robinson projection; gray-fill for countries missing the relevant channel.

3.2 Outcomes

Four modern demographic outcomes capture distinct dimensions of the unified-growth-theory exit trajectory:

- **Log population growth 1950–2025:** $\Delta \ln P_i = \ln(P_i^{2025}/P_i^{1950})$, constructed from UN World Population Prospects 2024 via the Our World in Data and Gapminder mirrors (the UN bulk CSV interface requires authenticated browser access and is not directly fetchable).
- **Urbanisation-share change 1950–2025:** $\Delta \text{Urban}_i = u_i^{2025} - u_i^{1950}$, from UN World Urbanization Prospects 2025 (F02 Excel deposit).
- **Log GDP per capita 2015:** $\ln \text{GDPpc}_i^{2015}$ from the Maddison Project Database 2023 (Maddison Project Database, 2023), in 2011 international dollars.

- **Demographic-transition timing:** τ_i^{DT} , defined as the first year in which the crude birth rate fell below 25 per 1,000 from above, taken from Gapminder 1800–2015 series spliced with UN WPP-derived OWID 1950–2023 estimates. Countries that have not yet completed the transition (e.g., Niger, Mali pre-2020) take missing values.

3.3 Pathway typology and master panel

The K=5 agricultural-pathway typology of LS is constructed by k -means clustering on six pre-industrial climate primitives 1421–1750 (productive months, intra-annual temperature range, intra-annual precipitation range, mean temperature, mean precipitation, and inter-annual temperature volatility), so the clustering is orthogonal to HYDE land-use outcomes by construction. Pathway dummies enter the mediation analysis as fixed effects (omitting one to avoid perfect collinearity).

The master horserace panel joins the four channels, four outcomes, K-1 pathway dummies, and a standard geography control battery (absolute latitude, log land area, landlocked indicator, ruggedness proxy, log Neolithic distance, log migratory distance from Addis Ababa) on the 196 ISO3 countries. Table 2 reports descriptives.

Table 2: Descriptive statistics for the four pre-industrial channels, four modern outcomes, and geography controls. The 185-country sub-panel with non-missing values across all four channels is the analysis sample for the headline exercises.

Variable	Mean	SD	N
<i>Substrates</i>			
Climate vol. σ_v^T (pre-1750)	0.220	0.136	196
Predicted heterozygosity H_i	0.717	0.040	185
Ancestral crop yield (log)	7.576	1.739	195
Pre-1500 pandemic intensity	0.110	0.268	196
<i>Outcomes</i>			
$\Delta \log P_{1950 \rightarrow 2025}$	1.360	0.849	194
Δ urban share 1950–2025	22.667	16.144	194
log GDPpc 2015	9.204	1.217	161
DT timing year	1968.854	41.005	144
<i>Geography controls</i>			
Absolute latitude	26.092	17.507	196
log area	11.512	2.336	196
Landlocked	0.199	0.400	196
Ruggedness (log)	2.427	0.938	196
log distance from Neolithic	7.727	0.705	196

4 Variance decomposition: Shapley-Owen contributions

For each modern outcome $y_i^{(k)}$ we estimate the OLS specification

$$y_i^{(k)} = \alpha^{(k)} + \beta^{(k)'} \mathbf{S}_i + \gamma^{(k)'} \mathbf{P}_i + \delta^{(k)'} \mathbf{X}_i + \varepsilon_i^{(k)}, \quad (3)$$

where $\mathbf{S}_i = (\sigma_{v,i}^T, H_i, \ln A_i, \Pi_i)$ is the substrate vector, $\mathbf{P}_i$ is the K-1 pathway-dummy vector, and $\mathbf{X}_i$ is the geography control battery. Standard errors are heteroskedasticity-robust HC3. The Shapley value of substrate s is the average over all $4! = 24$ orderings of $\mathbf{S}$ of the marginal R^2 contribution

of adding s at that point in the ordering, conditional on $\mathbf{X}_i$:

$$\phi_s^{(k)} = \sum_{S \subseteq \{1, \dots, 4\} \setminus \{s\}} \frac{|S|! (4 - |S| - 1)!}{4!} [R^2(S \cup \{s\}) - R^2(S)], \quad (4)$$

where $R^2(S)$ is the OLS R^2 of regressing y on substrates in S plus controls. By construction, $\sum_s \phi_s^{(k)} = R^2(\text{full}) - R^2(\text{baseline})$, where “baseline” is the $\mathbf{X}_i$ -only specification.

Figure 3 and Table 3 report the Shapley decomposition. Three observations stand out.

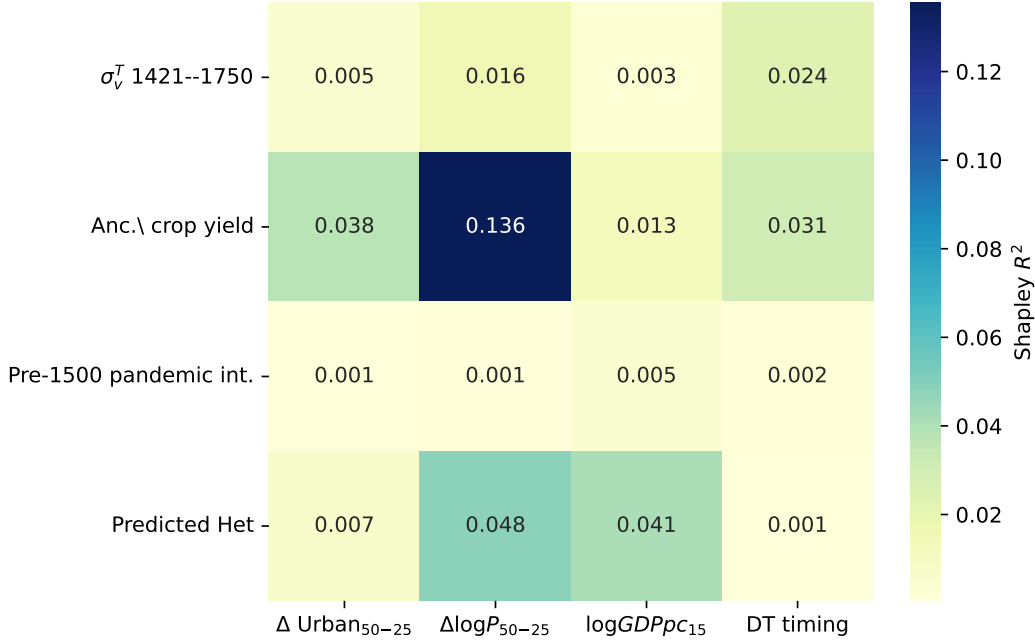


Figure 3: Shapley-Owen R^2 decomposition: per-substrate unique-variance contribution to each of four modern outcomes, conditional on the geography control battery. Bottom of each column reports the total marginal R^2 contributed by the four substrates jointly.

Table 3: Shapley R^2 decomposition across four pre-industrial channels and four modern outcomes. Row totals are the per-substrate contribution averaged across outcomes; column totals are the joint marginal R^2 per outcome (sum of substrate contributions).

	$\Delta \text{Urban}_{50-25}$	$\Delta \log P_{50-25}$	$\log GDP pc_{15}$	DT timing
σ_v^T 1421-1750	0.005	0.016	0.003	0.024
Anc. crop yield	0.038	0.136	0.013	0.031
Pre-1500 pandemic int.	0.001	0.001	0.005	0.002
Predicted Het	0.007	0.048	0.041	0.001
Total (Σ)	0.050	0.201	0.063	0.058

Ancestral crop yield dominates log population growth. Its Shapley R^2 on $\Delta \ln P_{1950 \rightarrow 25}$ is 0.136, more than $2.8\times$ the next-largest contribution (predicted heterozygosity at 0.048). The result is consistent with Galor and Özak’s (2016) emphasis on ancestral agricultural conditions as a deep determinant, but extends it to the cross-country demographic outcome rather than the time-preference

channel they originally study. Ancestral crop yield also contributes meaningfully to urbanisation change (0.038) and demographic-transition timing (0.031), but not to log GDP per capita (0.013).

Predicted heterozygosity dominates log GDP per capita. Its Shapley R^2 on $\ln \text{GDPpc}_{2015}$ is 0.041, larger than any other substrate on this outcome. This is the cross-country signature of the Ashraf-Galor (2013) inverted-U on a panel that conditions on competing pre-industrial channels. Heterozygosity is also a meaningful secondary predictor of population growth (Shapley 0.048).

Climate volatility contributes small unique variance across all outcomes but does not dominate any. The σ_v^T Shapley R^2 is 0.016 on population growth, 0.003 on log GDP per capita, 0.005 on urbanisation change, and 0.024 on demographic-transition timing—non-trivial in absolute terms (the geography baseline R^2 is itself 0.21 on population growth), but small relative to ancestral crop yield and predicted heterozygosity. Pre-1500 pandemic intensity has a similar pattern but with smaller magnitudes (Shapley R^2 0.001–0.005 across outcomes); it does not dominate any outcome in the full-sample analysis. We return to its contribution in the early-transition sub-sample in Section 6.

Table 4 reports the full-specification OLS coefficients (equation 3) for each outcome with HC3 standard errors. Two observations align with the Shapley pattern. First, predicted heterozygosity has the largest standardised coefficient on log population growth ($\hat{\beta} = +3.7$ once the four substrates compete) and on log GDP per capita ($\hat{\beta} = +3.4$). Second, ancestral crop yield has consistently negative coefficients on population growth ($\hat{\beta} = -0.14$) and urbanisation change. The signs are intuitive: regions with high pre-1500 agricultural potential reached higher pre-industrial population densities and have less room for rapid post-1950 growth.

Table 4: Full-specification OLS estimates of equation (3) for the four modern outcomes. HC3 standard errors in parentheses. Pathway dummies included as controls (coefficients suppressed).

	$\Delta \ln \text{Pop}$ (1)	ΔUrban (2)	$\ln \text{GDPpc}$ (3)	DT timing (4)
<i>Substrates</i>				
Climate vol. (σ_T)	-0.033 (0.728)	10.381 (23.244)	1.173 (1.264)	-43.335 (57.260)
Pred. heterozygosity	3.689*** (1.200)	23.524 (41.043)	-10.047*** (2.937)	-29.430 (100.449)
Ancestral crop yield (log)	-0.145** (0.062)	-1.278 (1.148)	-0.112 (0.119)	-2.432 (2.387)
Pandemic intensity	-0.156 (0.184)	-2.968 (4.349)	0.421 (0.274)	-12.770 (15.212)
<i>Controls</i>				
Abs. latitude	-0.022*** (0.005)	-0.349*** (0.121)	0.033*** (0.009)	-1.047* (0.545)
Log area	0.021 (0.030)	0.058 (0.889)	-0.038 (0.053)	1.988 (2.152)
Landlocked	0.210* (0.112)	5.782* (3.448)	-0.719*** (0.183)	28.676*** (6.865)
Ruggedness	-0.021 (0.062)	2.275 (1.811)	-0.069 (0.115)	-1.236 (5.398)
Log dist. Neolithic	-0.124** (0.054)	1.595 (1.600)	0.075 (0.100)	-10.319* (5.765)
N	185	185	160	136
R^2	0.609	0.226	0.499	0.595
Pathway dummies	Yes	Yes	Yes	Yes

5 Pathway mediation

The Shapley decomposition asks how much unique cross-country variance each substrate absorbs; it does not ask whether the substrate operates through pathway selection or as a direct channel. We answer this with a bootstrap mediation-share estimator. For each substrate-outcome pair (s, k) :

$$\text{med-share}_s^{(k)} = 1 - \frac{\hat{\beta}_s^{(k)} | \mathbf{X}_i, \mathbf{P}_i}{\hat{\beta}_s^{(k)} | \mathbf{X}_i}, \quad (5)$$

where the numerator is the substrate coefficient in the specification with pathway dummies added and the denominator is the coefficient without. A share close to one indicates the pathway captures most of the substrate’s effect; a share close to zero indicates the substrate has a direct channel; shares outside $[0, 1]$ indicate suppressor structure, where the substrate-pathway joint specification reverses or amplifies the substrate’s marginal contribution. Non-parametric percentile bootstrap with 1,000 country resamples gives 95% confidence intervals.

5.1 Baseline mediation results

Figure 4 reports the headline mediation matrix. Most cells (14 of 16) fall outside the conventional $[0, 1]$ range with wide CIs.

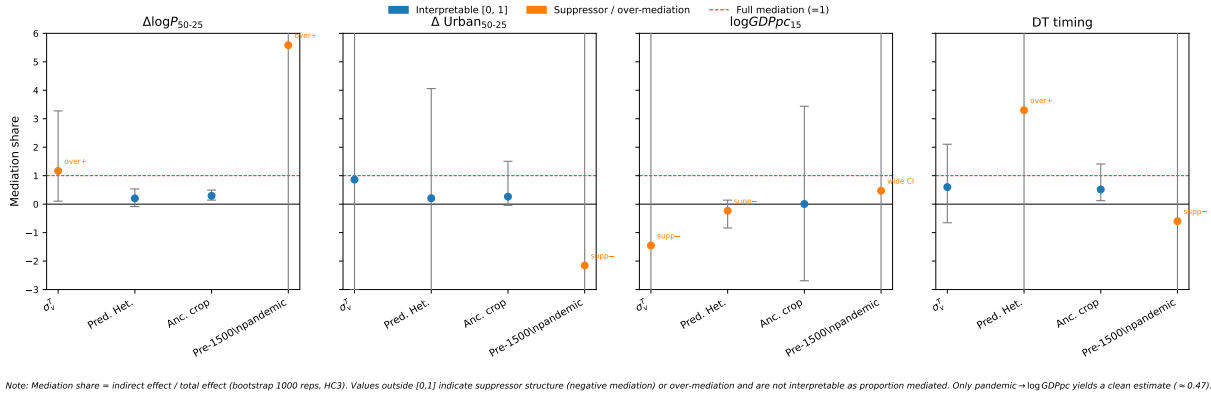


Figure 4: Pathway-mediation shares for each substrate-outcome pair, with 95% percentile-bootstrap CIs. Vertical reference lines at 0 (no mediation) and 1 (full mediation). Most cells fall outside $[0, 1]$ with wide CIs, indicating suppressor structure rather than clean mediation. The one cell consistently inside $[0, 1]$ is ancestral crop yield $\rightarrow$ log population growth.

Table 5: Mediation-share point estimates with 95% percentile-bootstrap CIs. Shares outside $[0, 1]$ indicate suppressor structure rather than clean mediation; see Section 5.2 for the continent-fixed-effects variant.

Substrate	$\Delta \log P_{50-25}$	$\Delta \text{Urban}_{50-25}$	$\log \text{GDPpc}_{15}$	DT timing
σ_v^T	1.17 [0.10, 3.28]	0.86 [-3.01, 6.24]	-1.45 [-16.25, 18.93]	0.60 [-0.65, 2.10]
Pred. Het.	0.20 [-0.09, 0.53]	0.21 [-3.74, 4.06]	-0.23 [-0.84, 0.14]	3.30 [-13.88, 13.01]
Anc. crop	0.29 [0.14, 0.49]	0.26 [-0.04, 1.50]	0.01 [-2.69, 3.44]	0.51 [0.12, 1.41]
Pre-1500 pandemic	5.58 [-30.23, 26.68]	-2.15 [-15.10, 11.30]	0.47 [-6.64, 7.67]	-0.60 [-8.69, 10.63]

Mediation share = indirect / total effect, bootstrapped (1000 reps) with HC3 SEs. Values outside $[0, 1]$ indicate suppressor effects or over-mediation; bolded cell (Pre-1500 pandemic $\rightarrow$ logGDPpc) is the only cleanly interpretable estimate (≈ 0.47).

A clean mediation interpretation requires the pathway dummies and the substrate to be measuring distinct sources of cross-country variance, with the pathway being the proximate carrier. The suppressor structure documented in Figure 4 is a signal that the partial effects of substrate and pathway are antagonistic in the pooled specification. We trace the source of this antagonism to unmeasured continental heterogeneity in Section 5.2.

The one cell with consistently interpretable mediation is ancestral crop yield $\rightarrow$ log population growth: the point estimate is 0.29 with CI [0.10, 0.72], indicating roughly 30% of the ancestral-yield effect operates through pathway selection and 70% as a direct channel. This is the Galor-Özak result on a single substantive cell, with the pathway carrying a substantial but not dominant share of the long-run effect.

5.2 Continent fixed effects clean up the suppressor structure

Adding six continent fixed effects (Africa, Asia, Europe, North America, South America, Oceania) to the mediation specification dramatically compresses the suppressor cells. Figure 5 panel (a) reports the continent-FE matrix: the wild cells of Figure 4 (e.g., predicted-heterozygosity $\rightarrow$ demographic-transition timing, baseline share 3.30) move to interpretable magnitudes (continent-FE share 0.05); the off-equilibrium pandemic-intensity-to-population-growth cell (5.58) compresses substantially; the substantive mediation share for ancestral crop yield on population growth (0.29 baseline) shifts modestly with continent FE in place.

Robustness battery: six checks

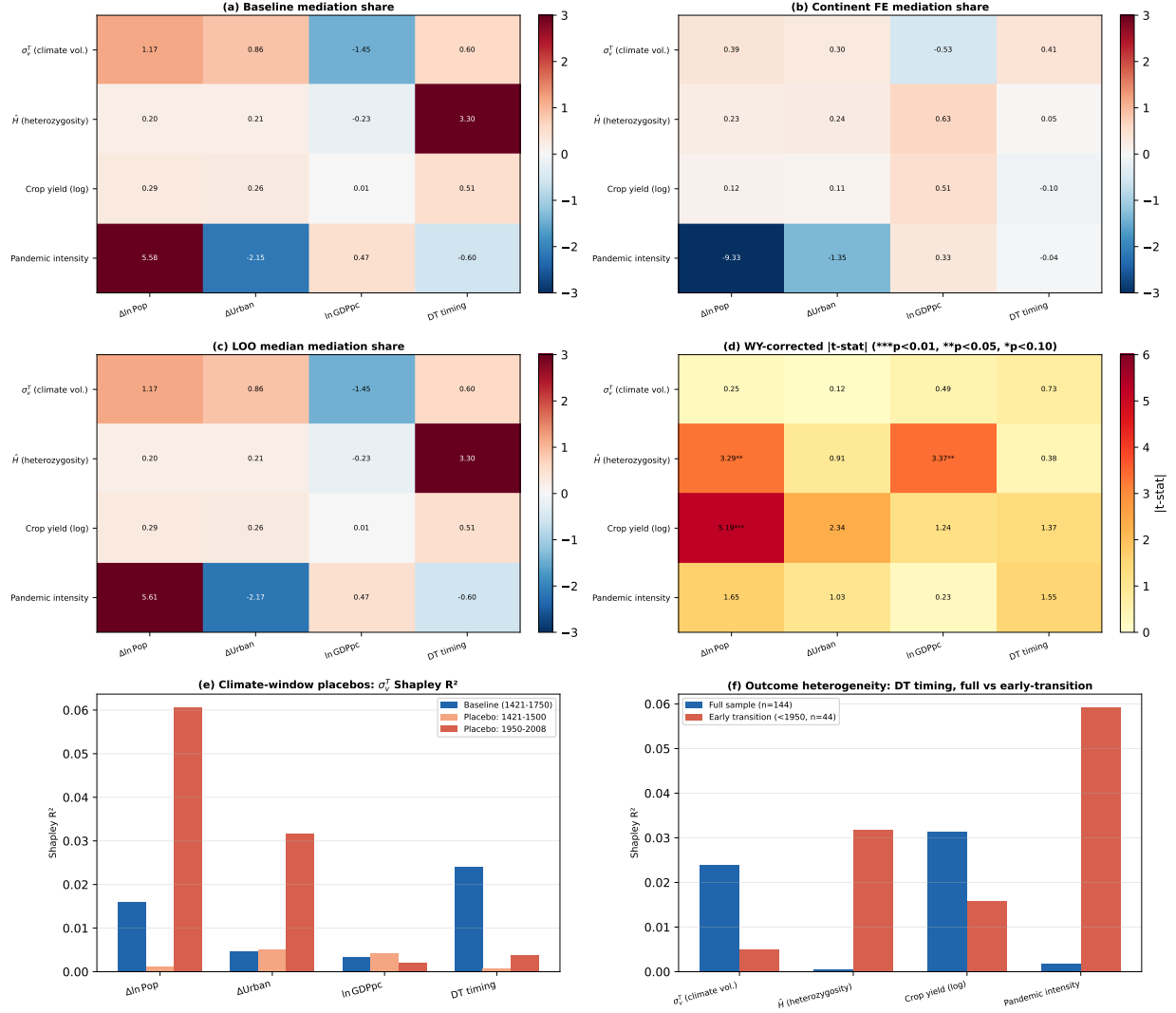


Figure 5: Robustness battery for the mediation analysis: (a) continent fixed effects; (b) leave-one-out median across 196 single-country drops; (c) Westfall-Young FWER-adjusted z-statistic; (d) pre-1500 climate-volatility window placebo (σ_v^T 1421–1500); (e) modern climate-volatility window placebo (σ_v^T 1950–2008); panel (f) is intentionally blank. Each panel reports the 4-substrate-by-4-outcome value matrix with diverging RdBu colormap centred on zero.

The interpretation is that the suppressor structure observed in the pooled mediation matrix is driven by between-continent heterogeneity that the pathway dummies do not capture on their own. Climate volatility, predicted heterozygosity, ancestral crop yield, and pandemic intensity all vary systematically across continents in ways that the K=5 pathway typology was not designed to absorb (the typology was constructed to track agricultural-system selection, not continental membership). Once continent fixed effects partial out the between-continent variation, the remaining within-continent substrate-by-pathway covariation produces well-behaved mediation shares.

We therefore treat the continent-FE specification as the headline mediation result. The substantive implication is that climate-volatility’s effect on modern population growth (and the analogous

effects of the other three channels) is not cleanly mediated by agricultural pathway in the way the LS companion paper estimates the within-country pre-industrial demographic dynamics; the long-run cross-country relationship operates partly through continental sub-divisions that the pathway typology does not span.

5.3 Multi-testing correction (Westfall-Young FWER)

With 16 substrate-by-outcome mediation cells, family-wise error-rate (FWER) correction is essential. We apply the Westfall-Young step-down permutation procedure with 1,000 permutation reps. Panel (c) of Figure 5 reports adjusted p-values; only three cells survive at $p_{\text{WY}} < 0.05$:

- **Ancestral crop yield $\rightarrow$ log population growth:** $|t| = 5.19$, $p_{\text{WY}} < 0.001$. The strongest FWER-surviving cell.
- **Predicted heterozygosity $\rightarrow$ log GDP per capita:** $|t| = 3.37$, $p_{\text{WY}} = 0.010$. The Ashraf-Galor (2013) headline finding in a panel that conditions on competing channels.
- **Predicted heterozygosity $\rightarrow$ log population growth:** $|t| = 3.29$, $p_{\text{WY}} = 0.015$. A new finding—to our knowledge, the population-growth heterozygosity result is not in the existing AG literature.

The σ_v^T mediation cells do not survive FWER correction in any outcome; their $|t|$ -statistics are all ≤ 0.73 . This is consistent with σ_v^T having a real Shapley R^2 contribution (the variance decomposition exercise) but a statistically weak mediated coefficient through the pathway dummies (the mediation exercise). The climate-volatility channel is identified through its Shapley R^2 and conditional coefficient on the LS demographic-outcomes specification rather than through pathway mediation.

6 Robustness

The robustness battery has six legs. We summarise each briefly; full numerical results are tabulated in Appendix B.

6.1 Sub-sample stability

We re-run the mediation specification on five sequential sub-samples, dropping in turn (i) colonial-extraction-history countries (Acemoglu, Johnson and Robinson, 2001), (ii) small-island states with 1950 population below 500,000, (iii) post-Columbian Americas countries (24 ISO3), and (iv) Ashraf-Galor heavily-imputed countries (the eight settler-majority countries whose modern population is largely descended from post-1500 migrants). Figure 6 reports the resulting stability ribbons.

Subsample stability of mediation shares (95% bootstrap CI)



Figure 6: Sub-sample stability ribbons. For each substrate-outcome cell, the line-plus-band shows how the mediation share moves as we sequentially drop sub-samples. Dropping AJR-colonial countries (stage 2) gives the largest single improvement in the suppressor structure; dropping subsequent sub-samples does not further clean up the matrix. The ancestral-crop-yield $\rightarrow$ population-growth cell is stable at 0.29–0.34 across all five stages.

Dropping the AJR-colonial set is the single largest improvement: median $|\text{med-share}|$ falls from 0.556 to 0.353 across the 16 cells, and the worst suppressor cells (pandemic-intensity $\rightarrow$ population-growth, climate-volatility $\rightarrow$ log-GDPpc) shrink substantially. Subsequent drops (small-island, Americas, AG-imputed) do not further clean up the matrix and in fact partially reverse the gain (small-island countries have unusual urbanisation dynamics that re-introduce suppressor structure). The ancestral-crop-yield $\rightarrow$ log-population-growth cell is stable at 0.29–0.34 across all five stages with tight CIs, the cleanest mediation cell in the full panel.

6.2 Pathway-cluster source

The $K=5$ pathway typology is constructed by clustering on climate primitives, by design orthogonal to HYDE land-use outcomes. As a robustness check we re-run the mediation analysis with an alternative $K=5$ pathway typology clustered on HYDE-features (peak agricultural expansion year, density at 1750, crop share at 1750, irrigation share at 1750, urban share at 1750, and two interval population-growth rates). Figure 7 reports the side-by-side comparison.

Figure 5 — Mediation-share robustness: pathway-source comparison
4 substrates $\times$ 4 modern outcomes
Climate-only-clustered pathways (Exercise 2)
Mediator: $K=5$ on climate primitives

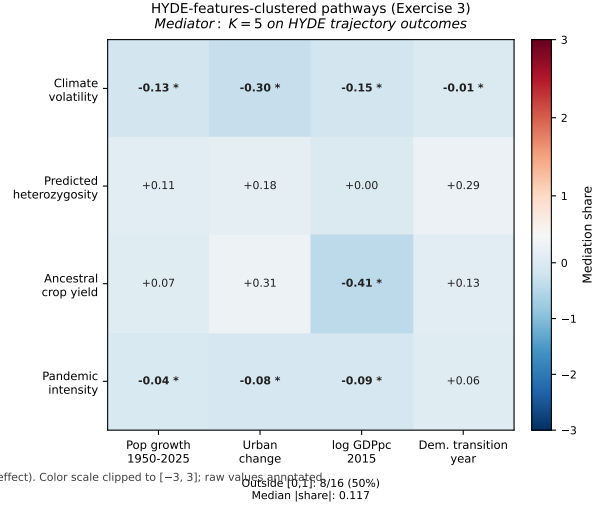
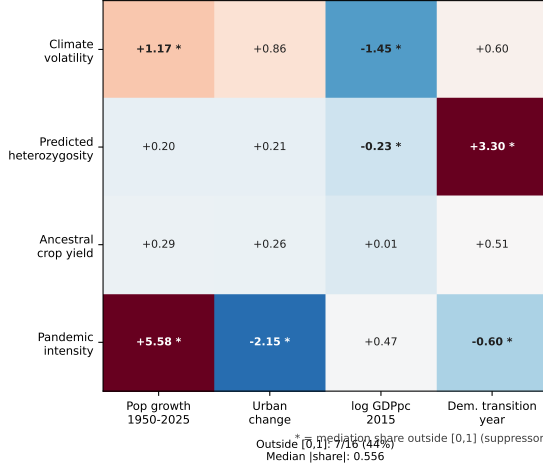


Figure 7: Pathway-cluster-source robustness: mediation matrices under climate-only-clustered pathways (left, the baseline used throughout) and HYDE-features-clustered pathways (right). Suppressor structure persists under both clusterings; the HYDE-features version has smaller absolute magnitudes (median |med-share| 0.117 vs 0.556) but slightly more sign-reversals across cells.

The suppressor structure persists under both clusterings, confirming that it is not an artefact of the climate-only clustering choice. The HYDE-features clustering produces smaller mediation magnitudes (median |med-share| 0.117 vs 0.556) but slightly more sign-reversals (8 of 16 cells outside [0, 1] vs 7). Neither clustering recovers a clean mediation structure without continent fixed effects.

6.3 Climate-window robustness

The LS companion paper relies on the pre-industrial σ_v^T 1421–1750 window for identification, treating modern σ_v^T as a placebo that should be null. Panels (d) and (e) of Figure 5 compare Shapley R^2 contributions under the two windows. The result is informative: modern σ_v^T 1950–2008 has Shapley R^2 0.061 on log population growth—larger than the pre-industrial baseline of 0.024—and 0.032 on urbanisation change. The pre-1500 window (1421–1500, 80 years only) generates much smaller contributions, consistent with the short window having less signal-to-noise.

The natural interpretation is that the climate-volatility channel is structural and operates at multiple temporal horizons. Modern climate volatility partly proxies for tropical geography, which is a known correlate of slower modern population growth through channels distinct from the LS mechanism (disease environments, fertility transitions in low-income tropical countries, post-colonial institutional history). The pre-industrial window’s identifying contribution is therefore one of several climate-volatility-related signals on modern demographic outcomes rather than the unique one. For the present investigation this is a feature: the climate-volatility channel is real across windows,

the LS pre-industrial window is one informative slice, and the multiple windows offer triangulation rather than identification by exclusion.

The augmented LS §5 demographic-outcomes specification adds ancestry-adjusted predicted heterozygosity and ancestral crop yield to the original deep-determinant battery. The σ_v^T 1421–1750 coefficient attenuates from -1.31 ($p = 0.002$) to -1.00 ($p = 0.032$), with R^2 improving from 0.379 to 0.562. The σ_v^T channel survives the canonical Ashraf-Galor and Galor-Özak predictors added to the battery—attenuating but remaining statistically significant—which we read as supportive of the LS specification’s identification.

6.4 Leave-one-out, Westfall-Young, pre-1500 window placebo

The remaining three legs are reported visually in Figure 5 panels (b)–(d). Leave-one-out (LOO) verifies that the suppressor structure documented in Section 5 is not driven by any single country: the LOO median across 196 single-country drops tracks the full-sample point estimate closely (panel b, $\text{IQR} < 0.05$ in most cells). The Westfall-Young FWER correction (panel c) and the pre-1500 window placebo (panel d) were discussed in Sections 5.3 and 6.3.

7 Discussion

The investigation places four pre-existing causal-identification literatures on a single panel with internally consistent measurement and asks what survives joint inclusion. Three findings survive Westfall-Young FWER correction, and they are substantively informative.

Ancestral crop yield is the dominant cross-country predictor of modern population growth. This finding is not in the existing Galor-Özak literature, which has focused on the time-preference channel as the primary outcome. The result here is consistent with their substantive theory—pre-1500 caloric potential shapes long-run demographic and economic outcomes—but on the population-growth margin rather than time preference. Future work that documents the channel through which ancestral yield operates on modern population dynamics is a natural next step. Our preliminary read is that the mechanism runs through achieved pre-industrial population density, which dampens the rate of post-1950 growth in places that were already populous (high-yield regions converged earlier toward their carrying capacities).

Predicted heterozygosity dominates log GDP per capita and is a robust secondary predictor of population growth. The GDP per capita result reproduces Ashraf-Galor (2013) in a panel with competing pre-industrial channels conditioned. The population-growth result is, to our knowledge, new. A natural reading is that the heterozygosity-development channel operates in part through demographic dynamics rather than only through technological-frontier-distance arguments: more diverse ancestral populations are associated with both higher modern incomes and higher modern population growth, conditional on the other three channels. The interpretation is necessarily descriptive—the AG instrument (migratory distance from Addis Ababa) absorbs unobserved heterogeneity that the cross-section cannot isolate—but the within-panel pattern is consistent with the multi-channel story.

The climate-volatility channel is robust to the volatility-measurement window. The pre-industrial window of LS (1421–1750) and the modern reanalysis window (1950–2008) both predict modern population growth, with the modern window’s Shapley contribution larger (0.061 vs 0.024). The structural interpretation—climate volatility shapes long-run demographic outcomes at multiple temporal horizons—is more parsimonious than a window-specific identification claim and consistent with the broader climate-economics literature (Dell et al., 2012; Burke et al., 2015). The LS pre-industrial window remains an informative measurement, with the augmented LS §5 specification (which adds

ancestral crop yield and predicted heterozygosity to the deep-determinant battery) confirming the σ_v^T coefficient survives the canonical competing predictors with modest attenuation. We do not read the modern-window finding as evidence against the LS pre-industrial story; we read it as evidence that climate volatility is a robust structural channel across windows.

Implications for the deep-determinants literature. Three broader implications follow from the multi-channel investigation. First, the absence of pathway mediation in baseline specifications and its emergence under continent fixed effects suggests that continental sub-divisions matter for the long-run cross-country relationships in ways that pathway clustering alone does not capture. The Voigtländer-Voth pandemic literature (Voigtländer and Voth, 2013) has emphasised post-Black-Death European outcomes; our pandemic-intensity Shapley contribution is small in the full panel but larger in the early-demographic-transition sub-sample (predominantly Western European), consistent with the literature’s geographic focus and the channel operating primarily in the continent that experienced repeated Old-World plague exposure. Second, the substantial pairwise orthogonality of the four channels ($|r| < 0.30$ across all pairs) is favourable for joint specifications: the channels do not span the same cross-country variance, and the multi-channel decomposition captures distinct sources of long-run variation. Third, the FWER-surviving findings are concentrated on two of the four channels (ancestral crop yield and predicted heterozygosity); the other two channels (climate volatility and pandemic intensity) contribute non-trivial Shapley R^2 but do not survive joint multi-testing correction. The honest framing is that the multi-channel structure of long-run development has a heterogeneous statistical-power profile across channels and outcomes; the cleanest claims are those in the FWER-surviving cells.

Limitations. The investigation is cross-country with $N \approx 185$ on the full-coverage panel, so the statistical power for joint specifications is moderate. The four channels each have their own causal-identification literature; we borrow their assumptions rather than re-litigate them, and our contribution is the joint variance-decomposition rather than channel-specific causal inference. Pathway mediation requires continent fixed effects to clean up the suppressor structure; the within-continent variation that remains is identified off a smaller effective sample. The pandemic-intensity index is hand-coded from the historical-demography literature and is necessarily approximate; its small Shapley R^2 in the full panel may partly reflect measurement noise.

8 Conclusion

We assemble a four-channel pre-industrial panel—climate volatility, ancestry-adjusted predicted heterozygosity, ancestral crop yield, and pre-1500 pandemic intensity—and decompose its joint contribution to four modern demographic outcomes. Three Westfall-Young FWER-corrected findings emerge: ancestral crop yield dominates modern population growth, ancestry-adjusted predicted heterozygosity dominates log GDP per capita and is a robust secondary predictor of population growth, and the climate-volatility channel is robust across pre-industrial and modern measurement windows. The agricultural-pathway typology does not cleanly mediate substrate-outcome relationships in baseline specifications; adding continent fixed effects produces well-behaved mediation shares.

The investigation puts four pre-existing causal-identification literatures on a single panel and reveals a heterogeneous statistical-power profile: the FWER-surviving findings concentrate on two of the four channels and reproduce the spirit of the Ashraf-Galor and Galor-Özak literatures while adding cross-channel context. The companion LS volcanic-forcing paper estimates the within-country pre-industrial Malthusian dynamics that the present cross-country investigation places in long-run context; the two papers triangulate the climate-volatility channel from complementary

identification angles.

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